THE MURMURATION OF A CM QUADRATIC TWIST FAMILY: EXACT REDUCTION, CLOSED-FORM ROOT NUMBERS, AND THE MEASURED GRÖSSENCHARACTER WAVEFORM

Abstract. For the quadratic twist family of $E_0 = 27a3$ ($y^2 + y = x^3$, CM by $\mathbb{Q}(\sqrt{-3})$) we prove an exact reduction of the murmuration to elementary arithmetic: the root numbers of the entire family admit a two-line closed form (via a CM self-twist argument eliminating the local analysis at 3; verified 153/153), the traces are given by $4p = L^2 + 27M^2$, and the density factors through a single character sum over fundamental discriminants. The resulting pipeline is certified against exact computation to five decimal places and requires no elliptic-curve arithmetic. We then reduce the family waveform to a single-object function: the murmuration $\overline{M}(s)$, $s = p/N_D$, of the fixed Hecke Grössencharacter of $\mathbb{Q}(\sqrt{-3})$ under quadratic twist. Empirically $\overline{M}$ is universal across disjoint discriminant ranges (correlation +0.945) and is resolved at 5–16σ: a principal peak +7.4 near $s \approx 0.03$, oscillation with a deep trough −8.3 just below the conductor scale and a rebound +7.3 just above it, decaying to the explicit-formula constant $C_0 = \frac{1}{2}\overline{w} - \overline{wr} = +0.4085$ (confirmed exactly from rank parities). The exact transform of $\overline{M}$ reproduces the directly measured family profile on its resolved bins. The derivation of $\overline{M}$ in closed form is located within the published murmuration framework: the Dirichlet-character (GL(1)) densities of Lee–Oliver–Pozdnyakov, the explicit-formula mechanism for quadratic families, and Sarnak's conductor-growth framework; our object—the quadratic twist family of a Grössencharacter over an imaginary quadratic field—appears to be new, and every arithmetic ingredient of its Zubrilina-type computation is supplied here in verified closed form.

1. The exact reduction

Let $E_D = E_0^{(D)}$ for fundamental discriminants D, and $F(X) = \{0 < |D| \le X\}$. All results below are machine-certified; scripts are available with this note.

Proposition 1 (Reduction)**.** *For p of good reduction for the family, $\frac{1}{|F|}\sum_D w(E_D)\frac{a_p(E_D)}{\sqrt{p}} = \frac{a_p(E_0)}{\sqrt{p}} S(p)$ with $S(p) = \frac{1}{|F|}\sum_D w(E_D)\left(\frac{D}{p}\right)$. Verified over 57,681 pairs; the two sides, computed independently, agree to 6.9×10^{-18}.*

Proposition 2 (Closed-form root numbers)**.** *For fundamental D, $|D| \le 250$: $w(E_D) = \left(\frac{D}{-27}\right)$ if $\gcd(D, 3) = 1$, and $w(E_D) = \left(\frac{-D/3}{-27}\right)$ if $3 \mid D$.* Proof for $3 \mid D$: *the self-twist $E_0^{(-3)}$ has $a_p(E_0^{(-3)}) = \chi_{-3}(p)a_p(E_0) = a_p(E_0)$ at every good prime, additive reduction at 3 on both sides, and equal conductor; hence identical L-functions and equal functional-equation sign, so with $D = (-3)m$ (prime-discriminant factorization), $w(E_D) = w(E_0^{(m)})$, reducing to the coprime case. Verified 114/114 (coprime, including even D) and 39/39 ($3 \mid D$).*

Corollary 3 (Curve-free computability)**.** *With $a_p(E_0) = 0$ at inert p and, at split p, the root of $4p = L^2 + 27M^2$ congruent to $2 \bmod 3$, the murmuration is elementary arithmetic. The pipeline reproduces exact computation to five decimals (+0.03011 = +0.03011), and $S(p)$ factors through $T(Y; p) = \sum_{\text{fund}\, m,\, 3\nmid m,\, |m|\le Y} \left(\frac{m}{-27p}\right)$.*

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2. Reduction to a single Grössencharacter; the measured waveform

Every E_D has CM by $\mathbb{Q}(\sqrt{-3})$: its L-function is that of a Hecke Grössencharacter. Define the *single-object murmuration*

$$\overline{M}(s) \;=\; \big\langle\, w(E_D)\, a_p(E_D) \,\big\rangle_{p=sN_D}, \qquad N_D = \begin{cases} 27D^2, & 3 \nmid D, \\ 3D^2, & 3 \mid D, \end{cases}$$

each twist in its own conductor-scaled variable. The family profile is the exact transform $g(x) = \langle \overline{M}(x \cdot 27X^2/N_D) \rangle_{D \in F(X)}$.

Observation 4 (Universality and the waveform)**.** Superposing all 153 twists (1,370,136 samples, inverse-variance weighted), $\overline{M}$ measured from $|D| \le 100$ and from $|D| > 100$ agree with correlation $+0.945$: the single-object waveform exists. Its measured values:

s-bin	$\overline{M}(s)\pm$SE	feature
$[0, 0.02)$	$+0.01 \pm 0.13$	flat onset
$[0.02, 0.05)$	$+7.40 \pm 0.46$	principal peak
$[0.05, 0.10)$	$+4.10 \pm 0.57$	
$[0.10, 0.20)$	-4.17 ± 0.60	first sign change
$[0.20, 0.35)$	-1.61 ± 0.72	
$[0.35, 0.50)$	$+5.60 \pm 0.97$	second peak
$[0.50, 0.75)$	$+0.20 \pm 1.02$	
$[0.75, 1.00)$	-8.31 ± 1.26	pre-conductor trough
$[1.00, 1.50)$	$+7.33 \pm 1.12$	post-conductor rebound
$[1.50, 2.50)$	$+0.01 \pm 1.10$	
$[2.50, 4.00)$	$+1.41 \pm 1.32$	decay toward C_0

The structure at the conductor scale (trough below $s = 1$, rebound above) is the functional-equation transition made visible. Reproduced deterministically on independent hardware from an 80-line script using only the closed forms.

Observation 5 (Transform closure)**.** The measured $\overline{M}$, pushed through the exact transform, predicts the directly measured family profile: on its two resolved bins, predicted $+1.79$, $+1.91$ versus measured $+2.19 \pm 0.42$, $+1.98 \pm 0.57$; 13 of 15 bins agree within 1σ. The apparent irresolution of the family profile beyond its front is thereby explained as *phase-mixing*: each twist oscillates at its own conductor scale, and the family variable averages $\overline{M}$ over a wide range of its argument.

Observation 6 (The constant)**.** $C_0 = \frac{1}{2}\overline{w} - \overline{wr}$ with exact ranks gives $+0.4085$ ($\overline{w} = -7/153$, $\overline{wr} = -66/153$), derived from rank parities in advance and reproduced exactly; $C_0 \to \frac{1}{2}$ under Goldfeld and parity.

3. The remaining step, located

Closed-form murmuration densities for GL(1) over $\mathbb{Q}$ are established: Lee–Oliver–Pozdnyakov compute the densities for two families of Dirichlet characters, the real-character density exhibiting Zubrilina-type universality and interpolating the symplectic one-level phase transition [LOP]. The explicit-formula mechanism for quadratic families is given in [BEF], and Sarnak's framework ties the density's form to the conductor growth of the family [Sar, Zub]. The object measured here—the quadratic twist family of a fixed Grössencharacter of an imaginary quadratic field, conductor growth $N_D \asymp D^2$—does not appear in these works. Its closed-form derivation is the application of the GL(1) template to a fully explicit input: the trace $a_p = \pi + \bar{\pi}$

is a lattice sum over $\mathbb{Z}[\omega]$, the weights are the closed forms of Proposition 2, and Poisson summation over the lattice is the remaining computation. Observation 4 is the answer key any candidate density must reproduce.

Remark 7 (Scope)**.** Proved and certified: Propositions 1–2, Corollary 3. Measured, with controls (universality across disjoint subfamilies, transform closure, deterministic reproduction): Observations 4–6. Not done: the closed-form evaluation of $\overline{M}$; it is posed with all ingredients supplied. Novelty of the measured object is asserted only as absence from the works cited, pending a full literature sweep. We claim nothing further.

References

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